

data that is backed up gets stored as files on the server's file system. No additional backup devices such as disk drives are needed (<https://www.boxbackup.org/>).

## Burp

Burp is a network backup and restore program. It attempts to reduce network traffic and the amount of space that is used by each backup. There are two independent backup protocols to choose from.

- Protocol 1: This optionally uses librsync. It is available in all Burp versions and is stable.
- Protocol 2: This uses variable length chunking for inline deduplication and sparse indexing on the server side.

Burp uses VSS (Volume Shadow Copy Service) to make snapshots when backing up Windows computers (<https://burp.grke.org/>).

## Duplicati

Duplicati is a backup client that securely stores encrypted, incremental, compressed backups on cloud storage servers and remote file servers. It provides various options and tweaks like filters, deletion rules as well as transfer and bandwidth options to run backups for specific purposes. It also supports the AES 256 encryption standards.

## Bareos

Bareos is a network and client-server based backup tool consisting of computer programs that empower IT administrators to manage the verification, backup and recovery of data through a network of computers. Bareos can also be made to run on a single computer, backing up various kinds of media (<https://www.bareos.org/en/>).

## BackupPC

BackupPC is a fully cross-platform tool that can run on all operating systems, and is designed for enterprise use. The provider supports full file compression and uses a small amount of disk space. Additionally, BackupPC doesn't need any

client-side software to run. The solution is freely available as open source on GitHub and SourceForge.

## Clonezilla

Clonezilla is open source backup software from Symantec Ghost Corporate Edition, based on DRBL (diskless remote boot Linux). The primary mechanism of the solution includes image partition and partial cloning, as well as bare metal backup and recovery udpcast. Two versions of Clonezilla are available — Clonezilla Live for single machine backup and restore, and Clonezilla SE for a server. The tool saves and restores only used blocks in the hard disk.

## Back in Time

Back in Time is backup software designed for Linux, and is inspired by the Flyback Project. The solution offers a command line client as well as a GUI, both written in Python. In order to perform backups, users specify where to store snapshots, what folders to back up and the frequency of the backups. The solution is licensed with GPLv2.

## Areca Backup

Areca Backup is an open source personal backup solution, released under the General Public License (GPL). Users may select a set of files or directories to back up, choose where and how they will be stored, and configure post-backup actions. The software also provides

encryption, compression and splitting functionalities. Areca allows customers to use advanced backup modes, such as delta backup, or produce a basic copy of their source files as a standard directory or Zip archive. The solution is compatible with Linux and Windows (<http://www.areca-backup.org/>).

## Disaster Recovery Linux Manager (DRLM)

DRLM provides disaster recovery to systems (desktops, servers, laptops, etc) using GNU/Linux, with standard tools and protocols. It is also used as a migration tool for all your system migrations. You can recover an entire system in a few minutes by just booting from the network, irrespective of whether your system is virtual or physical (<https://drlm.org/>).

## Efficient disaster recovery management

DR infrastructure must be aligned to meet recovery objectives for the various applications. An effective disaster management plan integrates the processes of reporting, monitoring and testing. It also provides workflow automation, and makes it a scalable and user-friendly solution for the organisation while not compromising on industry standards.

Recovery must be managed across virtual, physical and cloud infrastructure. It must incorporate advanced technologies but be perfectly in line with the needs and resources of the organisation. 

## References

- [1] <https://www.cio.com/article/3090892/8-ingredients-of-an-effective-disaster-recovery-plan.html>
- [2] <https://www.ibm.com/cloud/learn/backup-disaster-recovery#toc-what-are-b-qHZ-dz7m>
- [3] <https://netrixllc.com/blog/best-disaster-recovery-tools-2019/>

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